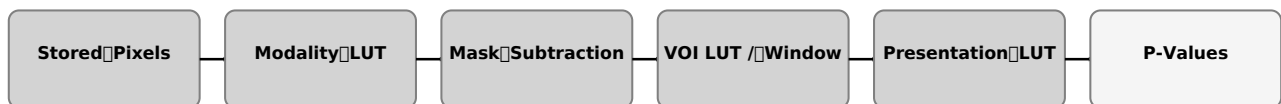


DICOM Pixel Data Transformation Pipeline

A deep implementation guide for Grayscale, PALETTE COLOR and TRUE COLOR RGB images — including Modality LUT, Mask Subtraction, VOI LUT, Presentation LUT, Palette Color LUT, ICC/PCS color management, shutter, annotations, spatial transformations, histograms and rendering

Audience: engineers implementing or reviewing a medical imaging viewer, DICOM rendering engine, GPU renderer, WebGL/Canvas pipeline or Presentation State interpreter.

Core idea: DICOM does not define one universal pixel-to-screen algorithm for every image. The pipeline depends fundamentally on whether the stored pixels represent a scalar grayscale measurement, a palette index, or already contain true color samples. The transformations therefore change from physical/modality meaning → display selection → presentation for grayscale, to index → color → color-managed display for palette color, and to RGB → color-managed display for true color.



Conceptual grayscale presentation pipeline

This guide uses the DICOM presentation model as the conceptual reference. The standard permits implementation details to differ as long as the required result/appearance is achieved; therefore a GPU renderer may fuse stages or use a different internal representation while preserving the specified semantics.

1. First principle: what does a stored pixel actually mean?

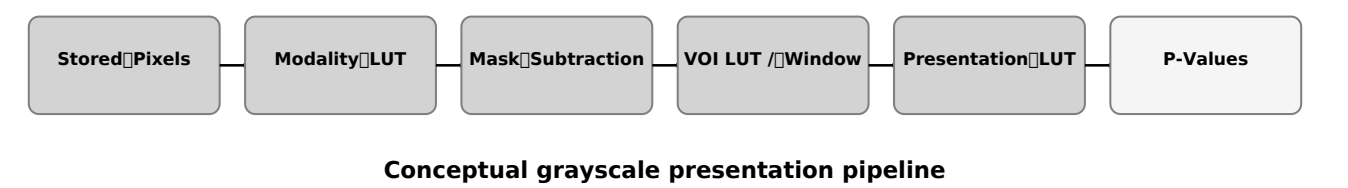
Before writing a renderer, ask one question: what semantic quantity does the stored sample represent? A CT stored sample may be a modality-dependent encoded value that becomes a Hounsfield Unit after the Modality LUT/rescale operation. A PALETTE COLOR stored sample is an index into a color table. A TRUE COLOR RGB sample is already a set of color components. These are not interchangeable representations.

Image representation	Stored sample means	Typical next semantic step	Can it be treated as scalar grayscale
MONOCHROME1 / MONOCHROME2	Intensity / modality-dependent sample	Modality LUT → VOI → Presentation	Yes
PALETTE COLOR	Index into R/G/B palette tables	Palette Color LUT → ICC/PCS	Not natively; index is not intensity
TRUE COLOR RGB	R/G/B color components	ICC/PCS color management when applicable	No
Grayscale → pseudo-color	Is Presentation LUT	Palette Color LUT → ICC/PCS	Yes, before pseudo-color LUT

Why this distinction matters: Modality LUT and VOI LUT have a semantic purpose. They are not generic 'image enhancement filters'. Applying them to arbitrary RGB channels would invent a modality interpretation that the DICOM object does not claim.

The DICOM standard's presentation model explicitly separates grayscale transformations from color transformations. Its color model includes true-color and pseudo-color paths and uses an ICC Input Profile to connect device-dependent color values to a Profile Connection Space (PCS). [\[cite\]turn2search0\[turn3search17\]](#)

2. The complete grayscale pipeline



Conceptual order: Stored grayscale values → Modality LUT → Mask Subtraction (when applicable) → VOI LUT / Window → Presentation LUT → P-Values → device/display function. Common spatial and annotation transformations are then applied according to the presentation model; the DICOM model places shutter and annotations around the spatial stage rather than treating them as intensity transforms.

DICOM PS3.4 identifies Modality LUT, Mask, VOI LUT and Presentation LUT as the grayscale transformation stages. The later common transformation model contains shutter, pre-spatial annotation, spatial transformation and post-spatial annotation. [\[cite\]turn3search2\[turn0search23\]](#)

3. Phase 1 — Stored Pixel Data

The decoder first reconstructs the stored samples from Pixel Data. This is still encoded image data, not necessarily a display-ready intensity. Bits Allocated, Bits Stored, High Bit, Pixel Representation, Photometric Interpretation and related attributes determine how those samples are interpreted.

Example: a signed 16-bit CT pixel value of 0 is not automatically 'black'. It is simply the stored sample. Its physical/modality meaning may be obtained by applying Rescale Slope and Rescale Intercept, e.g. $HU = \text{stored} \times \text{slope} + \text{intercept}$.

Engineering rule: never use the display histogram to infer modality meaning. Build the modality-domain values first when the object specifies a Modality LUT or Rescale operation.

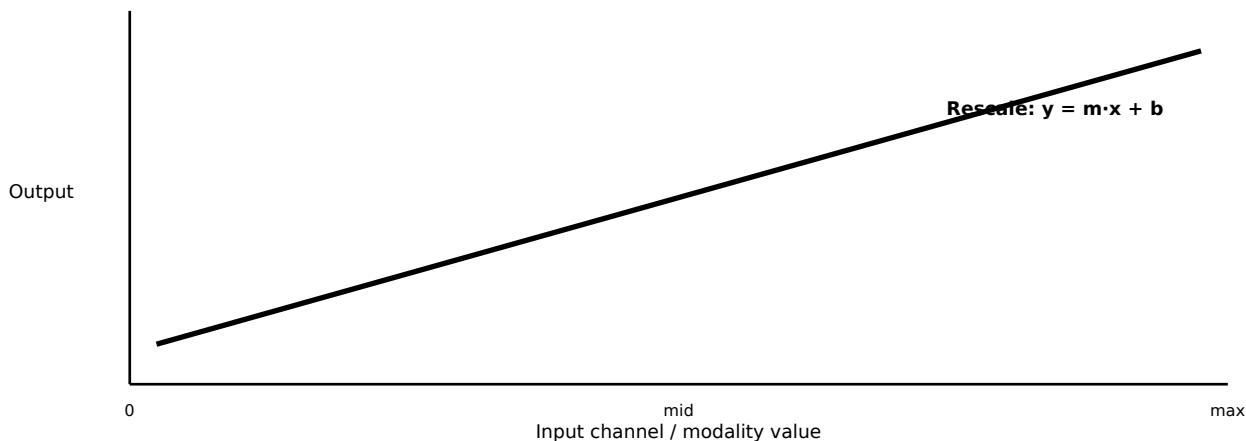
4. Phase 2 — Modality LUT: why it is fundamental

The Modality LUT is the bridge between how a modality encoded a measurement and what that measurement means. DICOM describes it as transforming manufacturer-dependent pixel values into values meaningful for the modality and manufacturer-independent, such as Hounsfield Numbers for CT or Optical Density for film digitizers. It applies only to grayscale values in the grayscale pipeline. [\[cite\]turn3search17\]](#)

There are two common representations of the Modality LUT operation: an explicit LUT, or a linear Rescale Slope/Intercept transformation. DICOM requires an unambiguous modality transformation for the input to subsequent grayscale stages. [cite]turn0search19[

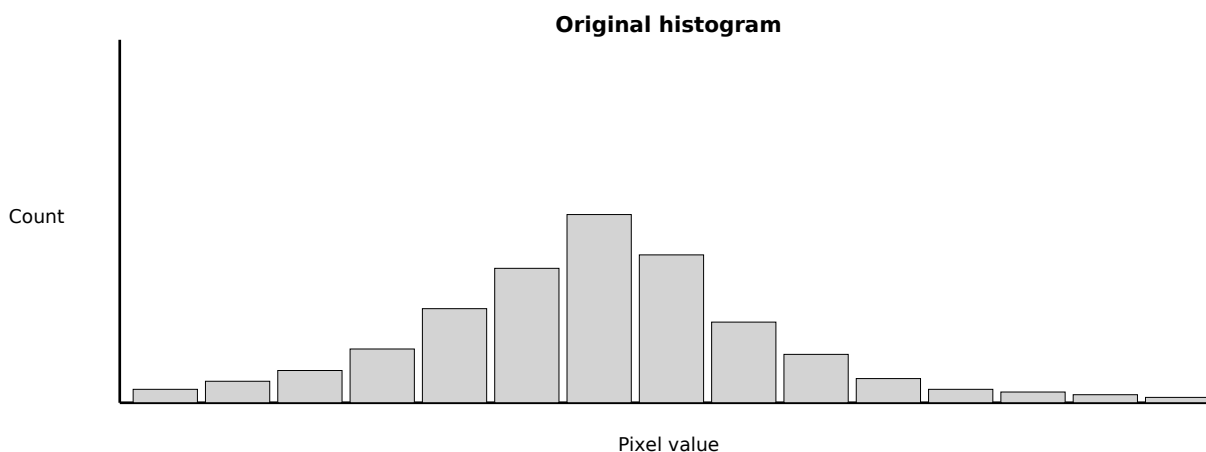
Linear form: $\text{ModalityValue} = \text{StoredValue} \times \text{RescaleSlope} + \text{RescaleIntercept}$

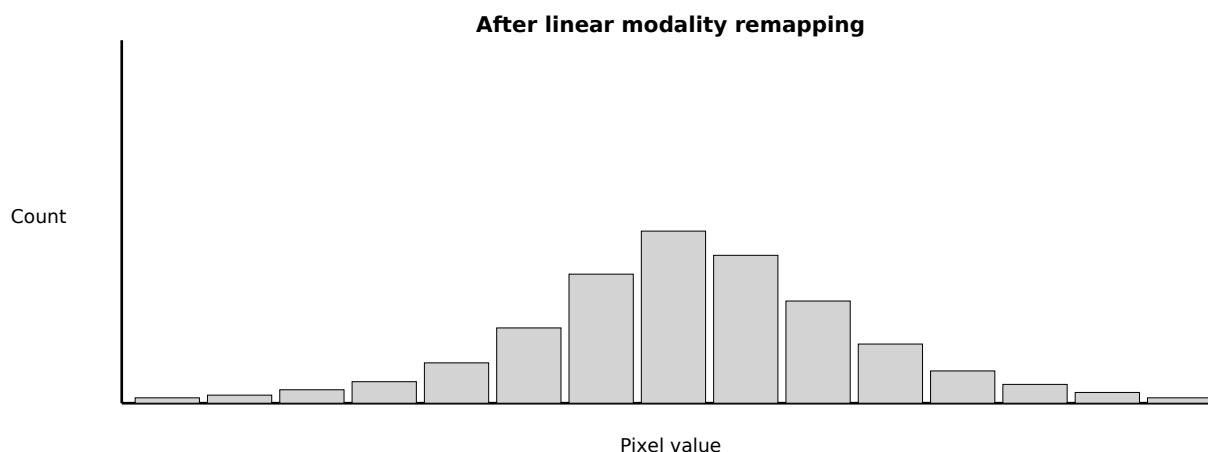
Example: Stored = 500, slope = 1, intercept = -1024 → modality value = -524 HU. Stored = 1500 → 476 HU. The same stored integer therefore acquires a physically meaningful interpretation only after this step.



The transformation changes the numerical coordinate system. It does not merely make the picture 'brighter' or 'darker'. A slope changes scale; an intercept shifts the origin. Consequently, window center/width values are meaningful in the modality domain, not blindly in the original stored-code domain.

5. Modality LUT and histogram — what technically changes?





For a linear transformation $y = m \cdot x + b$ with $m > 0$, the ordering of pixels is preserved, but their numerical positions move and their spacing changes. If you histogram in the new numerical domain, the histogram's x-axis is transformed. The count of pixels represented by a region is not magically changed; the labels/coordinates of the bins change. With a negative slope, ordering is reversed.

Important: a histogram is not an intrinsic object independent of its domain. A CT histogram in stored values and a histogram in HU are different coordinate representations of the same pixels. If you compare them without accounting for the transform, you can reach incorrect conclusions.

6. Phase 3 — Mask Subtraction

Mask subtraction is not a universal grayscale operation. It is primarily associated with angiographic/XA/XRF presentation workflows where a mask frame can be subtracted from another frame to emphasize contrast-filled vessels or suppress background anatomy. The exact mask operation is defined by the relevant presentation state and modality-specific attributes.

Conceptually, for two registered frames A and M, a simple subtraction looks like: $D(x,y) = A(x,y) - M(x,y)$. Real DICOM XA/XRF processing can include mask selection, pixel shifts, intensity relationships, LUT-related behavior and mask visibility percentage; it is therefore more than a generic image-difference filter. [\[cite turn3search18\]](#)

Why after Modality LUT? The presentation model describes the mask stage between Modality LUT and VOI for the general grayscale transformation path. This gives the subtraction operation a defined modality-value context. In specialized XA/XRF pipelines, the standard defines a modality-specific sequence that may include mask and edge-enhancement processing. [\[cite turn3search18\]](#)

Applicability: Native RGB and native PALETTE COLOR images do not normally undergo grayscale mask subtraction merely because they are images. A modality/presentation-state that explicitly defines a specialized operation is a separate case.

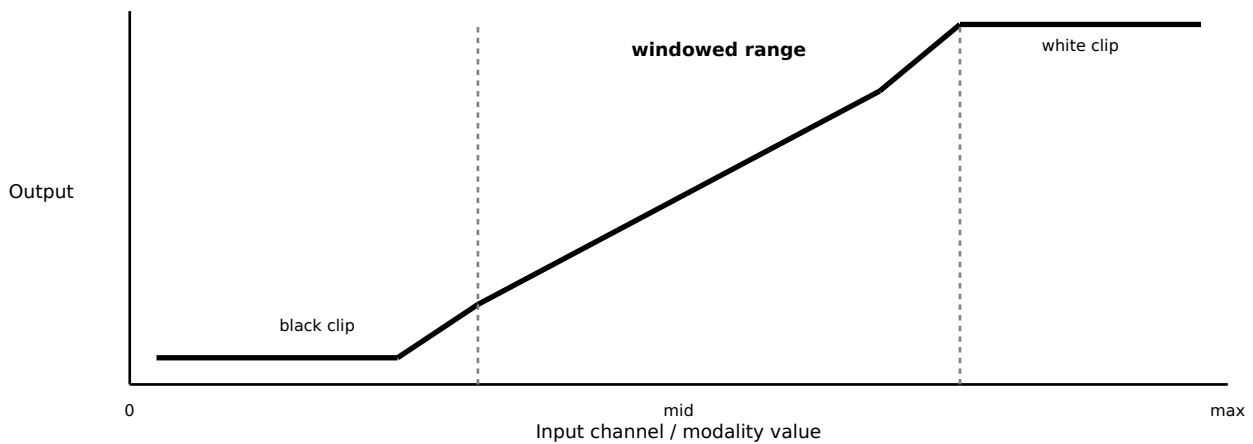
7. Phase 4 — VOI LUT / Window Center & Window Width

VOI means Value of Interest. The VOI stage takes modality-domain values and selects the range that is useful for the observer/application. In CT, for example, a lung window, bone window and soft-tissue window can emphasize very different value ranges from the same underlying dataset.

DICOM permits an explicit VOI LUT or a Window Center/Window Width transformation. For the standard LINEAR window function, the transformation maps a selected input interval to the output range and clips values below/above the interval. DICOM explicitly defines the formula and boundary behavior. [\[cite turn1search4\]](#)

Conceptual simplified equation: $y = \text{clamp}((x - (C - W/2)) / W, 0, 1)$. The exact DICOM definition includes the -0.5 and $(W-1)$ terms to define consistent discrete behavior; do not substitute the simplified equation when bit-exact conformance is required.

[\[cite turn1search4\]](#)

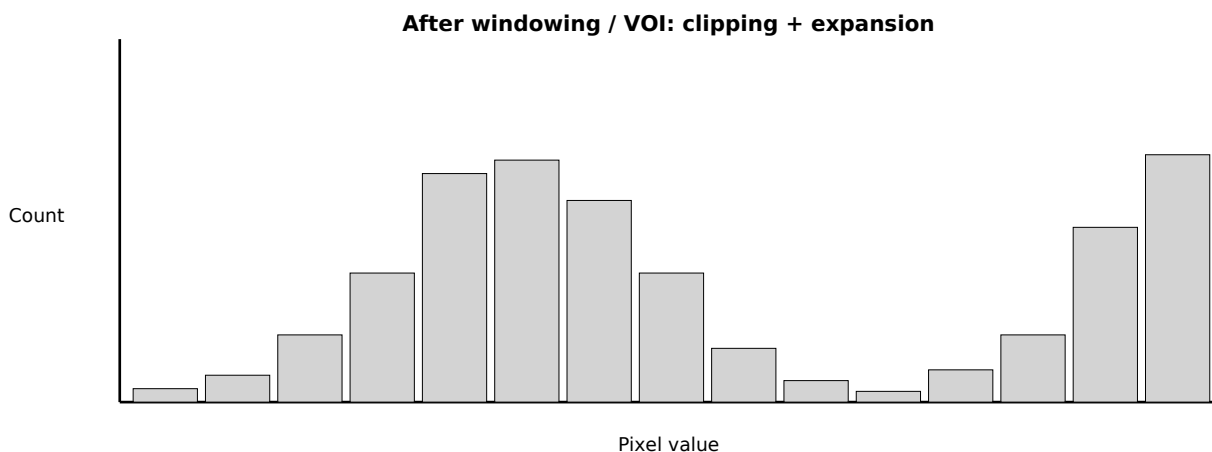


8. What VOI actually changes — numerically and functionally

Numerically: VOI usually compresses a selected modality-value interval into the available display range and collapses values outside that interval into the low/high output limits. It therefore changes the numerical pixel values presented downstream.

Functionally: VOI changes what information is visually emphasized. It does not change the underlying DICOM Pixel Data. Switching from a brain window to a bone window is changing the mapping of the same source measurements, not rewriting the acquisition.

Histogram consequence: windowing typically creates large accumulations at the output minimum and maximum because all values below the lower bound map to the same low value and all values above the upper bound map to the same high value. The interior range is stretched. This is why a post-VOI histogram can have strong spikes at black and white even when the source histogram does not.



This is also why histogram statistics must state which pipeline stage they describe: stored-pixel histogram, modality-value histogram, VOI-output histogram, or final display-value histogram.

9. Why VOI is so important in medical imaging

Medical images often contain a much larger numerical dynamic range than a display can represent simultaneously. VOI is the mechanism that selects the clinically useful portion of that range. A CT study may contain air, lung, fat, soft tissue, bone and metal across a broad HU range. A single fixed mapping would either waste display codes or compress diagnostically useful differences.

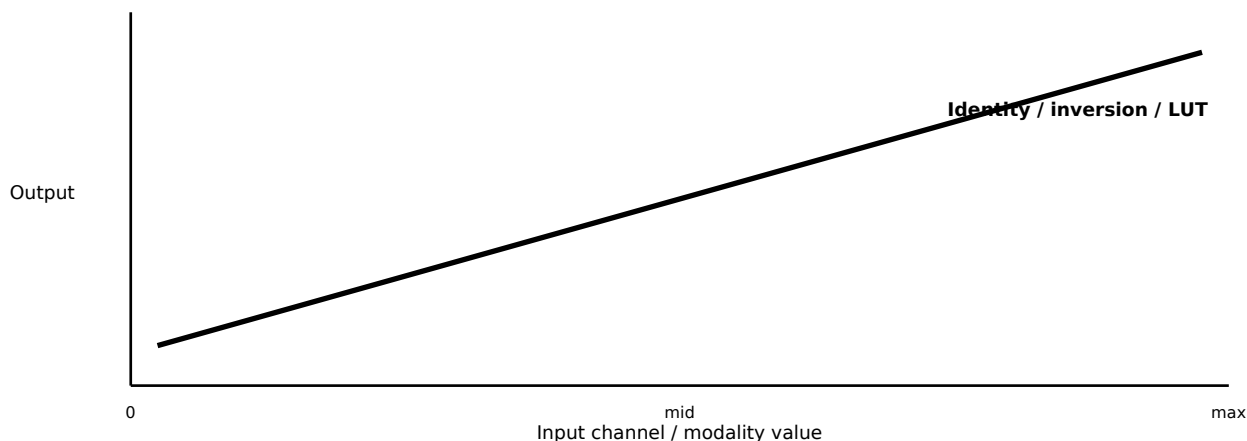
Window width controls the size of the emphasized range; window center controls where that range is positioned. A narrow window increases local visual contrast within that range. A wide window preserves a broader range but

produces less visual separation per unit of modality value.

Key distinction: VOI is not equivalent to generic CSS brightness/contrast. VOI is a modality/application-aware mapping whose input is normally the modality-domain value. CSS contrast is a display-space affine operation and does not know that a CT value represents HU.

10. Phase 5 — Presentation LUT (PLUT)

The Presentation LUT is the final grayscale presentation mapping before P-Values. Its purpose is to translate the post-VOI grayscale values into P-Values, which are device-independent presentation values intended to provide a common input to standardized display systems. DICOM describes P-Values as approximately perceptually related and independent of the characteristics of a particular display device. [cite]turn1search2[turn1search5]



Why is PLUT required? Because selecting a clinical value range is not the same thing as defining how the resulting grayscale codes become standardized presentation values. The Presentation LUT provides the final presentation-domain mapping, including polarity-related behavior where required. It may be an identity transformation when the preceding stage already produces appropriate P-Values.

11. Types / forms of Presentation LUT

Form	Concept	Typical purpose
Identity	Output = input	Use when preceding grayscale pipeline already produces P-Values / no further m
Inverse	High becomes low and low becomes high	Supports polarity/inversion semantics such as MONOCHROME1 presentation.
Explicit LUT	Input code → LUT entry	Defines a specific presentation mapping, e.g. modality/application/display requir

DICOM also defines Presentation LUT Shape semantics for identity/inverse behavior. For example, an identity shape indicates output is already in P-Values; an inverse shape can convert the polarity so the displayed relationship is appropriate. [cite]turn3search6[

Do not think of PLUT as 'another window/level'. VOI answers which modality values should occupy the display range? PLUT answers how the resulting grayscale presentation values are represented as standardized display values?

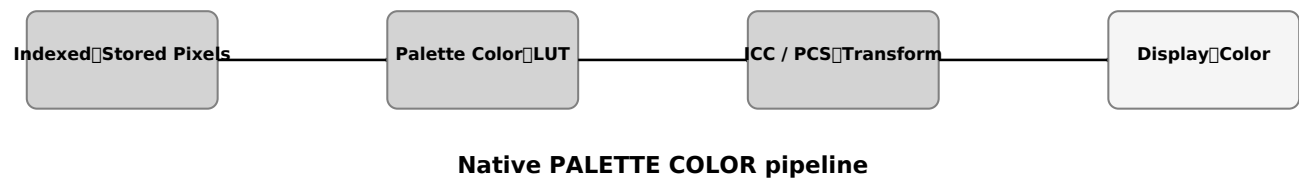
12. P-Values and GSDF — the final grayscale contract

P-Values form the boundary between the DICOM image/presentation pipeline and the standardized display system. PS3.14 defines the Grayscale Standard Display Function (GSDF), which relates P-Values to luminance using a perceptual model based on just-noticeable differences. The goal is consistent grayscale presentation across standardized display systems, not identical physical luminance from every consumer display. [cite]turn1search0[turn1search3[

Pipeline mental model: Stored code → modality meaning → clinically selected range → standardized presentation value → display luminance.

This is why the PLUT/P-Value concept is important even though a modern GPU may simply render an 8-bit texture. The numerical texture format alone does not tell you whether those codes are stored values, modality values, VOI values or P-Values.

13. Native PALETTE COLOR DICOM images



In a native PALETTE COLOR image, each stored sample is an index. The index selects an entry from three palette tables: Red, Green and Blue. The palette LUT therefore expands one stored sample into three color components.

DICOM's Palette Color LUT Module defines red, green and blue palette data and descriptors for PALETTE COLOR images. [cite]turn0search22[

Critical semantic difference: an index value of 100 does not mean '100 units of intensity'. It means 'look up entry 100'. Applying CT-style Modality LUT or window/level to the index would normally be semantically wrong because the index is not a physical modality measurement.

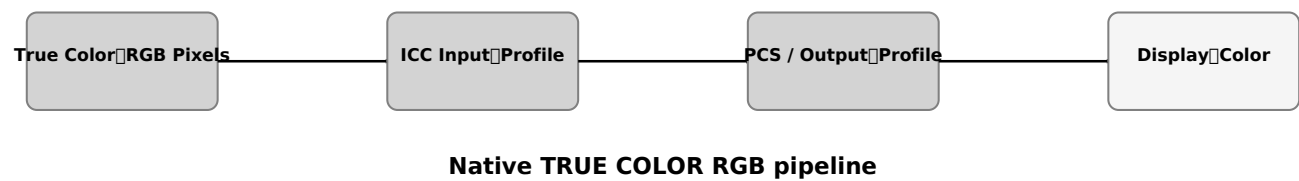
After palette expansion, the resulting RGB values may undergo color management through an ICC Input Profile / Profile Connection Space transformation. DICOM explicitly includes both pseudo-color and true-color images in the color transformation model. [cite]turn2search0[

14. Why Modality LUT and VOI are generally not applied to native PALETTE COLOR

Stage	Native PALETTE COLOR	Reason
Modality LUT	Normally no	Stored sample is a palette index, not a modality scalar.
Mask subtraction	Normally no	Requires a defined grayscale/modality-value subtraction workflow; color index subtraction is not applicable.
VOI / Window	Normally no	Windowing a palette index would alter palette selection, not select a modality-value range.
Palette Color LUT	Yes	This is the fundamental index → RGB operation.
ICC / PCS	Potentially yes	Color management converts source color characteristics into a device-independent color space.
Presentation LUT	Not the normal native color	Grayscale PLUT is part of grayscale presentation; color follows the color/PCS model.
Shutter / spatial / annotations	Yes, where specified	These operate on the presentation geometry/content rather than assuming grayscale intensity.

There is an important exception-like scenario: a grayscale source may be transformed into pseudo-color by a Presentation State. In that case Modality LUT and VOI can operate first on the grayscale source, and the resulting values are then mapped through a Palette Color LUT and ICC profile. DICOM's transformation model explicitly shows this branch. [cite]turn3search19[turn3search20[

15. Native TRUE COLOR RGB DICOM images



For native true-color RGB, the stored samples already represent color components. The central problem is therefore not 'convert modality codes to a scalar physical quantity and window them'; it is 'interpret the color correctly and render it consistently'.

The DICOM color model applies a Profile Connection Space transformation to true-color images. An ICC Input Profile describes the color characteristics of the source values; the rendering system can convert them to a device-independent PCS and then to an output/display profile. [cite]turn2search0[

Why no Modality LUT? Because R, G and B are not three measurements that become a single modality scalar through the grayscale modality transformation. Treating each channel as though it were a CT HU-like measurement would change the intended color representation.

16. ICC profiles: what changes and what does not

An ICC transformation is fundamentally about color meaning and reproducibility, not simply brightness. The source RGB triplet is interpreted according to the source profile and converted to a device-independent Profile Connection Space. The output side can then use a display/output profile to obtain suitable device values.

Conceptually: Source RGB → ICC Input Profile → PCS (e.g. Lab/XYZ) → Output/Display Profile → Display RGB. The DICOM standard states that the exact color-management implementation is outside the standard, but the conceptual goal is color consistency. [cite]turn2search0[

Important: two identical RGB triplets can represent different colors if they belong to different color spaces/profiles. Conversely, two different device RGB triplets can be converted to approximately the same PCS color. This is why copying raw RGB values without respecting their color profile can change perceived color.

DICOM states that the ICC profile in a Color, Pseudo-Color or Blended Presentation State takes precedence over an ICC profile in the image when both are applicable. [cite]turn2search0[

17. Palette LUT vs ICC Profile — do not confuse them

Mechanism	Input	Output	Primary job
Palette Color LUT	Single index	RGB triplet	Decode indexed/pseudo-color representation.
ICC Input Profile	Source RGB/color values	PCS values	Describe/convert source color characteristics.
Display/output profile	PCS	Device-specific RGB/device values	Adapt standardized color to the display.
VOI LUT	Scalar modality values	Display-range scalar	Select/expand clinically relevant value range.
Presentation LUT	Post-VOI grayscale	P-Values	Prepare grayscale presentation values for standardized display.

A Palette LUT answers which RGB color does this index mean? An ICC profile answers what color does this RGB triplet mean in a device-independent color space? They solve different problems.

18. Shutter transformation

A shutter excludes a perimeter/region from display and can replace the excluded region with a specified gray level. DICOM also defines bitmap-based shuttering for arbitrary excluded areas. Importantly, the standard notes that annotations follow the shutter operation, so annotations in shuttered regions remain visible.

[cite]turn0search23[

It is not a VOI operation. A shutter changes spatial visibility, not the meaning of the underlying pixel value. Therefore it should not be implemented as 'set pixels outside ROI to black' in the source image buffer unless the renderer understands that this is presentation-only behavior.

19. Annotations: pre-spatial vs post-spatial

DICOM distinguishes annotations that are defined in image pixel space from annotations defined in displayed-area space. This distinction determines whether the annotation should move/rotate/scale with the image or remain anchored to the displayed viewport.

Annotation type	Coordinate space	Effect of image rotation/zoom	Typical examples
Pre-spatial annotation	Image pixel space	Transforms with the image	Pixel-space overlay, graphic annotation attached to image
Post-spatial annotation	Displayed-area space	Remains associated with display area	Viewport text, labels, presentation-state graphics

The common DICOM model explicitly places pre-spatial annotation before the spatial transformation and post-spatial annotation after it. [cite]turn0search23[

20. Spatial transformation and Displayed Area

Spatial transformation changes the image's presentation orientation. The DICOM model supports rotations of 90°, 180° and 270° clockwise and horizontal flipping; arbitrary-angle rotation is not defined by that Spatial Transformation mechanism. Displayed Area selection can crop/zoom the selected region, with interpolation/subsampling implementation-dependent. [cite]turn0search23[

Important architectural separation: intensity transforms change pixel values; spatial transforms change where those pixels appear. Mixing these concerns makes a renderer difficult to reason about and can create incorrect annotations or windowing behavior.

21. Applicability matrix — all major stages

Stage	Grayscale	Native PALETTE COLOR	Native TRUE COLOR RGB	Why
Decode Pixel Data	✓	✓	✓	All images need samples reconstructed.
Modality LUT / Rescale	✓	Normally ✗	✗	Converts grayscale stored codes to modality-specific values.
Mask subtraction	Conditional	Normally ✗	Normally ✗	Specialized grayscale/angiographic operations.
VOI LUT / Window	✓	Normally ✗	✗	Selects a scalar value-of-interest range.
Presentation LUT / P-Values	✓	Not normal color path	Not normal color path	Defines grayscale presentation values.
Palette Color LUT	Conditional branch	✓	✗	Maps scalar/index values to RGB.
ICC / PCS	Conditional for pseudo-color when applicable	✓ when applicable	✓ when applicable	Color consistency / color-space conversion.
Shutter	✓	✓	✓	Presentation-space exclusion.
Pre-spatial annotation	✓	✓	✓	Pixel-space graphics/overlays.
Spatial transformation	✓	✓	✓	Orientation/displayed-area operations.
Post-spatial annotation	✓	✓	✓	Display-area-relative graphics/text.

Do not read '✗' as 'the DICOM standard can never define such a transformation anywhere'. It means the transformation is not the normal semantic stage for that native pixel representation. Specialized Presentation States, blending workflows, pseudo-color transformations and modality-specific pipelines can introduce additional paths.

22. One source, three pipelines — side-by-side reasoning

Question	Grayscale	Palette Color	True Color RGB
What is a pixel?	Scalar measurement/code	Color-table index	R/G/B samples
Need physical/modality normalization?	Often Yes	No for the index itself	No scalar normalization
Window/level?	Yes	No for native index	No
How is color introduced?	Optional pseudo-color LUT	Palette LUT	Already present
Color consistency problem?	Only if converted to color	Yes	Yes
Final color management?	If output becomes color	ICC/PCS	ICC/PCS
Clinical window changes raw data?	No	No	No
Main rendering challenge	Dynamic range + polarity + GSDF	Correct index/LUT + color profile	Correct color space/profile

23. Histogram engineering — a stage-aware model

A robust viewer should conceptually distinguish at least four histograms: stored histogram, modality histogram, VOI-output histogram, and display/output histogram. They answer different questions.

Histogram	What it tells you	Common use
Stored pixel histogram	Distribution of encoded samples	Debugging decoder/bit depth/sign issues.
Modality histogram	Distribution in HU, OD or other modality units	Clinical range analysis, auto-windowing, QA.
VOI histogram	Distribution after window mapping/clipping	Understanding clipping and display utilization.
Final display histogram	Distribution after presentation/color conversion	Renderer/display diagnostics; not a substitute for modality histogram.

Auto-windowing should usually be based on the modality domain when the goal is to choose a clinically meaningful window. Doing percentile analysis directly on stored codes can be valid only when the mapping and desired behavior are explicitly understood.

24. A worked CT example from stored code to display

Assume an illustrative CT pixel has $\text{StoredValue} = 1400$, $\text{RescaleSlope} = 1$ and $\text{RescaleIntercept} = -1024$. The modality value is 376 HU.

Now choose Window Center $C = 40$ HU and Window Width $W = 400$ HU. The selected interval is approximately -160 to $+240$ HU in the simplified interpretation. A value of 376 lies above the upper boundary, so its VOI output becomes the high display value. Another pixel at 40 HU lands near the middle of the VOI output range.

If the viewer changes to a bone window, the same stored pixels are not re-decoded differently. The modality values remain the same; only the VOI mapping changes. This is the conceptual basis of interactive window/level controls.

25. Why transformation order matters

These operations are not generally commutative. A modality transform followed by VOI is not equivalent to VOI followed by modality transform, because the window parameters are defined in a particular value domain. Similarly, converting a grayscale image to RGB and then trying to apply grayscale windowing to RGB channels is not equivalent to windowing first and then applying a color LUT.

Think in terms of domains: Stored Domain $\rightarrow$ Modality Domain $\rightarrow$ VOI Domain $\rightarrow$ Presentation Domain $\rightarrow$ Display/Color Domain. Each stage expects a particular semantic input.

A common implementation bug is to optimize by moving a mathematically similar operation earlier in the pipeline without checking whether the operation's parameters are defined in the same domain. A renderer can be mathematically fast and still be clinically wrong.

26. GPU implementation architecture

A high-performance viewer should not necessarily materialize a new full-resolution CPU buffer after every stage. Many operations can be represented as uniforms/LUT textures and composed in a fragment shader. For example, a grayscale shader can fetch the stored value, apply slope/intercept, perform VOI, apply presentation mapping and finally write a display texture.

A practical conceptual shader sequence is:

stored $\rightarrow$ modality(stored) $\rightarrow$ mask/derived value $\rightarrow$ voi(modality) $\rightarrow$ presentation(voi) $\rightarrow$ output

For palette color, the shader can instead perform: stored index $\rightarrow$ palette texture lookup $\rightarrow$ color-management transform/output. For RGB, it can fetch RGB directly and perform the applicable color-management conversion.

Optimization rule: preserve semantic stage boundaries even if implementation stages are fused. For example, fusing Modality LUT + VOI into one shader is safe if the fused function produces exactly the same result as the two conceptual stages.

27. Debugging a DICOM rendering pipeline

Symptom	Likely stage to inspect	What to log
CT window appears wrong	Modality / VOI	Stored value, slope, intercept, modality value, WC, WW.
MONOCHROME1 appears inverted	Presentation/polarity	Photometric Interpretation, Presentation LUT Shape, PLUT.
Palette colors look wrong	Palette LUT / ICC	Index, RGB LUT entry, ICC profile presence and selection.
RGB image has color cast	Color profile management	Photometric Interpretation, Color Space, ICC profile.
Annotations rotate incorrectly	Spatial / annotation stage	Pre/post spatial coordinate system.
Shutter hides annotations	Ordering bug	Shutter must precede relevant annotation stage.
Auto-window inconsistent	Histogram domain	Confirm histogram is calculated in modality domain.
GPU and CPU results differ	Precision / LUT / rounding	Compare every stage, especially VOI boundaries and LUT indexing.

Best diagnostic technique: capture one pixel and trace it through every stage. Then trace a small histogram/ROI. Do not debug only by looking at the final image.

28. Pixel-trace worksheet

Stage	Example value	Expected representation
Stored Pixel	1400	Signed/unsigned stored sample
Modality LUT	376	HU / modality unit
Mask	376 or derived subtraction result	Modality-domain result
VOI	high/interior/low output	Display-range scalar
Presentation LUT	P-Value	Device-independent grayscale presentation
Shutter	unchanged inside / replacement outside	Presentation geometry
Spatial transform	rotated/flipped coordinates	Display geometry
Annotation	graphic/text value	Overlay/presentation content
Display	luminance/color	Device output

29. Common implementation mistakes

1. Windowing stored values instead of modality values when the DICOM object specifies Rescale Slope/Intercept. 2. Applying VOI to RGB channels as if they were a grayscale modality value. 3. Applying a grayscale Presentation LUT to native RGB simply because the pipeline has a 'PLUT step'. 4. Ignoring ICC profiles and assuming raw RGB numbers are universally meaningful. 5. Using a post-VOI histogram to derive a clinical window. 6. Rotating the image buffer but not the pixel-space annotations. 7. Applying shutter after post-spatial annotations and accidentally hiding annotations that should remain visible. 8. Changing the conceptual order during optimization without proving equivalence.

30. The most useful mental model for engineers

Grayscale: What does this number mean? → Modality LUT. Which part of the meaningful range do I want to see? → VOI. How do I express the result as standardized presentation values? → Presentation LUT / P-Values. Where should it appear? → shutter, annotations, spatial/displayed area.

Palette Color: Which color does this index select? → Palette Color LUT. What does this RGB color mean and how should it appear on another display? → ICC/PCS color management. Spatial and annotation transformations remain presentation concerns.

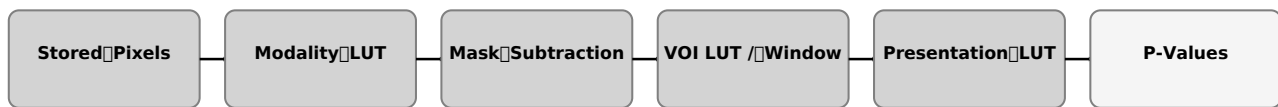
True Color RGB: The image is already color. Do not invent a grayscale modality pipeline. Interpret the color representation and apply the applicable color-management path, then handle presentation geometry and annotations.

One sentence: Modality LUT changes meaning; VOI changes what part of that meaning is visible; Presentation LUT changes grayscale presentation representation; Palette LUT changes index into color; ICC changes color-space/device interpretation; spatial/annotation stages change where and how the result is presented.

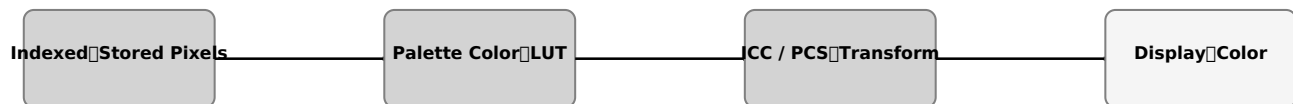
31. Reference architecture for a viewer renderer

Layer	Recommended responsibility
DICOM decode	Transfer Syntax, Pixel Data, bit depth, sign, planar configuration, frame extraction.
Semantic normalization	Photometric Interpretation and representation-specific interpretation.
Grayscale pipeline	Modality LUT → mask where defined → VOI → Presentation LUT/P-Values.
Palette pipeline	Index → Palette Color LUT → ICC/PCS where applicable.
RGB pipeline	RGB interpretation → ICC/PCS where applicable.
Presentation geometry	Shutter → pre-spatial annotation → spatial/displayed-area transformation → post-spatial annotation, according to
GPU renderer	Fuse stages only when semantic equivalence is preserved.
QA/debug	Per-pixel trace + stage-specific histograms + reference images.

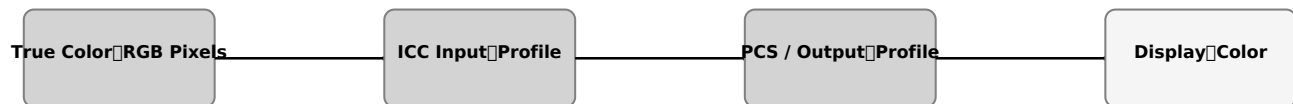
32. Final comparison diagram



Conceptual grayscale presentation pipeline



Native PALETTE COLOR pipeline



Native TRUE COLOR RGB pipeline

The three pipelines share a presentation/geometry world, but their pixel-semantic paths are different. This is the single most important architectural distinction to preserve in a DICOM viewer.

33. Standards-based notes and references

The principal normative references used for this guide are the DICOM Standard, especially PS3.3 for image modules/attributes, PS3.4 Section N for Softcopy Presentation State transformation models, and PS3.14 for the Grayscale Standard Display Function. The current DICOM site provides the 2026c release and current PS3 documents. [cite]turn0search8[turn0search2]

- DICOM PS3.4, Section N.2 — Pixel Transformation Sequence: grayscale, color, common spatial/annotation transformations and blending. [cite]turn3search2[turn2search0]
- DICOM PS3.3, Palette Color LUT Module — palette descriptors and RGB palette data. [cite]turn0search22
- DICOM PS3.4 / VOI definition — Window Center/Width and VOI LUT behavior, including exact linear windowing boundaries. [cite]turn1search4
- DICOM PS3.14 — Grayscale Standard Display Function and P-Value concepts. [cite]turn1search0[turn1search3]
- DICOM color transformation model — ICC Input Profile, Profile Connection Space and color consistency. [cite]turn2search0
- DICOM educational material illustrating grayscale and pseudo-color transformation models. [cite]turn3search19

Conformance caution: this guide intentionally uses simplified equations in explanatory sections. For bit-exact conformance, implement the exact DICOM-defined LUT indexing, windowing boundary rules, output ranges, signedness, descriptors, rounding and Photometric Interpretation behavior specified by the applicable IOD and presentation state.